

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	HDI Data Input	Allows users to enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita.	User Interface	Done	High
2	Data Validation	Validates user input to ensure correct and complete data before prediction.	Backend	Done	High
3	Data Preprocessing	Cleans and preprocesses the HDI dataset for machine learning.	Data Processing	Done	High
4	Machine Learning Model	Trains and predicts the Human Development Index (HDI) category.	ML Module	Done	High
5	HDI Category Prediction	Displays the predicted HDI category (Very High, High, Medium, or Low)	Prediction Module	Done	High
6	Flask Web Application	Provides a web-based interface for entering data and viewing prediction results.	Web Application	Done	Medium
7	Model Storage	Saves and loads the trained machine learning model for future predictions.	Model Management	Done	Medium
8	Data Visualization & Reports	Displays charts or reports to help users analyze development indicators and prediction results.	Model Management	In Progress	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	7
Core / Must-Have Features	6 (Data Input, Validation, Preprocessing, ML Model, Prediction, Flask Web Application)
Additional / Nice-to-Have Features	2 (Model Storage, Data Visualization & Reports)
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	HDI Data Input Form, Prediction Result Display
2	Backend / Logic	3	Data Validation, Data Preprocessing, HDI Prediction
3	Database / Storage	2	HDI Dataset (CSV), Trained Model (.pkl using Joblib)
4	API / Integration	1	Flask API for Prediction Requests
5	Security / Authentication	1	Input Validation and Error Handling (<i>Basic security; authentication can be added later if required</i>)